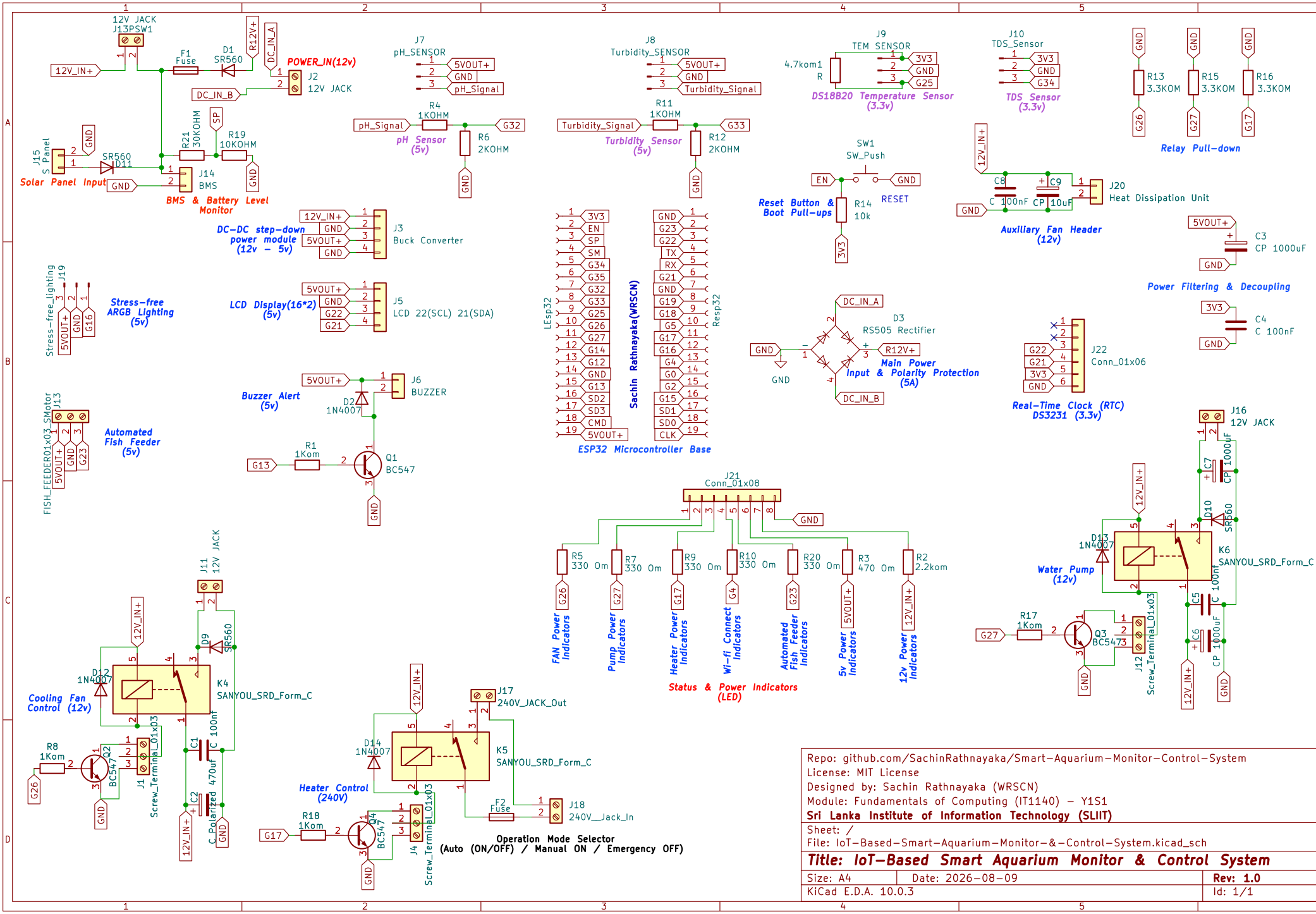
12V JACK
J13PSW1
12V_IN+
F1
Fuse
D1
SR560
R12V+
DC_IN_A
POWER_IN(12v)
J2
12V JACK
DC_IN_B
J15
S Panel
Solar Panel Input
SR560
D11
GND
R21
30KOHM
SP
R19
10KOHM
J14
BMS
BMS & Battery Level Monitor
J7
pH_SENSOR
5VOUT+
GND
pH_Signal
R4
1KOHM
pH Sensor (5v)
R6
2KOHM
G32
J8
Turbidity_SENSOR
Turbidity_Signal
R11
1KOHM
Turbidity Sensor (5v)
R12
2KOHM
G33
J9
TEM SENSOR
4.7kom1
R
3V3
G25
DS18B20 Temperature Sensor (3.3v)
J10
TDS_Sensor
G34
TDS Sensor (3.3v)
R13
3.3KOM
R15
3.3KOM
R16
3.3KOM
G26
G27
G17
Relay Pull-down
SW1
SW_Push
EN
RESET
R14
10k
Reset Button & Boot Pull-ups
C8
C 100nF
C9
CP 10uF
J20
Heat Dissipation Unit
Auxiliary Fan Header (12v)
C3
CP 1000uF
Power Filtering & Decoupling
C4
C 100nF
J19
Stress-free_lighting
Stress-free ARGB Lighting (5v)
DC-DC step-down power module (12v - 5v)
J3
Buck Converter
LCD Display(16*2) (5v)
J5
LCD 22(SCL) 21(SDA)
LEsp32
Resp32
Sachin Rathnayaka(WRSCN)
ESP32 Microcontroller Base
D3
RS505 Rectifier
Main Power Input & Polarity Protection (5A)
J22
Conn_01x06
Real-Time Clock (RTC) DS3231 (3.3v)
J13
FISH_FEEDER01x03_SMotor
Automated Fish Feeder (5v)
Buzzer Alert (5v)
J6
BUZZER
D2
1N4007
R1
1Kom
Q1
BC547
J21
Conn_01x08
R5
330 Om
R7
330 Om
R9
330 Om
R10
330 Om
R20
330 Om
R3
470 Om
R2
2.2kom
FAN Power Indicators
Pump Power Indicators
Heater Power Indicators
Wi-fi Connect Indicators
Automated Fish Feeder Indicators
5v Power Indicators
12v Power Indicators
Status & Power Indicators (LED)
J16
12V JACK
C7
CP 1000uF
D10
SR560
D13
1N4007
K6
SANYOU_SRD_Form_C
Water Pump (12v)
C5
C 100nF
C6
CP 1000uF
R17
1Kom
Q3
BC547
J12
Screw_Terminal_01x03
J11
12V JACK
D9
SR560
D12
1N4007
K4
SANYOU_SRD_Form_C
Cooling Fan Control (12v)
R8
1Kom
Q2
BC547
J1
Screw_Terminal_01x03
C1
C 100nF
C2
C_Polarized 470uF
D14
1N4007
K5
SANYOU_SRD_Form_C
J17
240V_JACK_Out
Heater Control (240V)
R18
1Kom
Q4
BC547
J4
Screw_Terminal_01x03
F2
Fuse
J18
240V__Jack_In
Operation Mode Selector
(Auto (ON/OFF) / Manual ON / Emergency OFF)
Repo: github.com/SachinRathnayaka/Smart-Aquarium-Monitor-Control-System
License: MIT License
Designed by: Sachin Rathnayaka (WRSCN)
Module: Fundamentals of Computing (IT1140) - Y1S1
Sri Lanka Institute of Information Technology (SLIIT)
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